Chapter 16, Problem 10RQ

Problem

A single-input, single-output system is described by the state model as:

$$\dot{x}_1 = 2x_1 - x_2 + 3z$$
$$\dot{x}_2 = -4x_2 - z$$
$$y = 3x_1 - 2x_2 + z$$

Which of the following matrices is incorrect?

(a) $\mathbf{A} = \begin{bmatrix} 2 & -1 \\ 0 & -4 \end{bmatrix}$

(b) $\mathbf{B} = \begin{bmatrix} 3 \\ -1 \end{bmatrix}$

(c) $\mathbf{C} = [3 \quad -2]$

(d) **D** = **0**

Step-by-step solution

Step 1 of 1

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Write a single-input, single-output system is described by state model.

$$\begin{aligned} \dot{x}_1 &= 2x_1 - x_2 + 3z \\ \dot{x}_2 &= -4x_2 - z \qquad \text{...... (1)} \\ y &= 3x_1 - 2x_2 + z \end{aligned}$$

Write the state equations.

$$\begin{aligned} \dot{\mathbf{x}} &= \mathbf{A}x + \mathbf{B}z \\ \mathbf{y} &= \mathbf{C}x + \mathbf{D}z \end{aligned} \qquad \text{...... (2)}$$

Compare equation (1) with equation (2) to put in matrix form.

$$\mathbf{A} = \begin{bmatrix} 2 & -1 \\ 0 & -4 \end{bmatrix}$$
$$\mathbf{B} = \begin{bmatrix} 3 \\ -1 \end{bmatrix}$$
$$\mathbf{C} = \begin{bmatrix} 3 & -2 \end{bmatrix}$$
$$\mathbf{D} = \begin{bmatrix} 1 \end{bmatrix}$$

Thus, the correct option is $\boxed{\text{d}}$.